



Experiments to determine how likely an outcome is

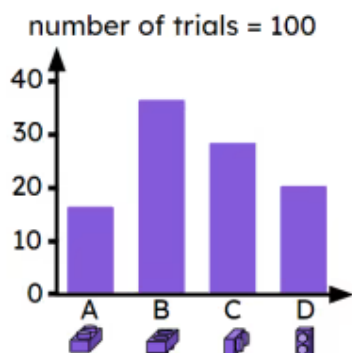
1 _____ is a repetition of a trial multiple times in order to observe how often each outcome occurs. Tick **1** correct answer

- ☐ An event
- ☒ An experiment
- ☐ A sample space

2 The _____ an experiment contains, the more confident you can be about any conclusions made based on its results. Tick **1** correct answer

- ☐ fewer trials
- ☐ more possible outcomes
- ☒ more trials

3 During an experiment, a toy brick is dropped 100 times and its landing position is noted. Based on the results in the bar chart, which way is the brick least likely to land when it is dropped? Tick **1** correct answer



- ☒ A - with connectors up
- ☐ B - with connectors down
- ☐ C - on one of its long faces
- ☐ D - on one of its short faces



- 4 Sofia is planning experiment involving dropping slices of buttered toast to see which way around it is most likely to land. Which number of trials would produce the most reliable conclusion? Tick **1** correct answer

- ☐ 1
☐ 2
☐ 10
☐ 50
☒ 100

- 5 Jacob conducts an experiment where he flips a coin 10 times to check whether landing 'heads' is equally likely as landing 'tails'. How could this experiment be improved? Tick **1** correct answer

- ☐ use a fewer number of trials
☒ use a greater number of trials
☐ asking someone else to flip the coin
☐ the experiment can not be improved

- 6 Jacob flips a coin 10 times. It lands 'heads' 3 times and 'tails' 7 times. Which statement is true? Tick **1** correct answer

- ☐ The coin is definitely biased because it landed on tails more than heads.
☐ The coin is definitely not biased because results can differ due to chance.
☒ You cannot reliably conclude whether the coin is biased based on so few trials.

